Year 13 Further Maths - Polar coordinates

Similar questions - Areas, loops and optimisation

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) The curve $r = \sin 3\theta$ for $0 \leqslant \theta \leqslant \frac{\pi}{3}$ forms a single loop. Find the exact area enclosed by the loop. **[4]**

(b) Find the exact area enclosed by the cardioid $r = 4(1+\cos\theta)$, $0 \leqslant \theta \leqslant 2\pi$. **[4]**

2 The curve C has polar equation $r = 2(1+\cos 2\theta)$ for $0 \leqslant \theta < 2\pi$ and consists of two identical loops, one either side of the pole.

(a) Show that a Cartesian equation of C is $\left(x^2+y^2\right)^{3/2} = 4x^2$. **[3]**

(b) Show that the x-axis is a line of symmetry. **[2]**

(c) In this part you must show detailed reasoning. Find the exact area of each loop. **[5]**

3 An art gallery floor region R is the region lying inside the curve $C_1 : r = 6$ and outside the curve $C_2 : r = 4\cosh\theta$, for $-\theta_0 \leqslant \theta \leqslant \theta_0$, where the curves meet at $\theta = \pm\theta_0$. Distances are in metres. Each tin of paint covers 0.5 m^2.

(a) Show that $\theta_0 = \cosh^{-1} 1.5$. **[2]**

(b) In this part you must show detailed reasoning. Determine the smallest number of tins needed to paint R completely. **[7]**

4 For $k > 0$, the curve C_k has polar equation $r = k(1+\cos\theta) + \dfrac{6}{k}$ for $0 \leqslant \theta < 2\pi$. It encloses an area A_k.

(a) Show that $A_k = \pi\left(\frac{3}{2}k^2 + 12 + \dfrac{36}{k^2}\right)$. **[5]**

(b) Determine, in exact form, the value of k for which A_k is a minimum, and the exact minimum area. **[4]**

Question 1

(a) $A = \frac{1}{2}\int_0^{\pi/3} \sin^2 3\theta \, \mathrm{d}\theta = \frac{1}{4}\int_0^{\pi/3} (1-\cos 6\theta)\, \mathrm{d}\theta = \frac{1}{4}\cdot\frac{\pi}{3} = \dfrac{\pi}{12}$

(b) $A = \frac{1}{2}\int_0^{2\pi} 16(1+\cos\theta)^2 \mathrm{d}\theta = 8\int_0^{2\pi} \left(\frac{3}{2} + 2\cos\theta + \frac{1}{2}\cos 2\theta\right) \mathrm{d}\theta = 8 \times 3\pi = 24\pi$

Question 2

(a) $r = 2(1+\cos 2\theta) = 4\cos^2\theta$, so $r^3 = 4r^2\cos^2\theta = 4x^2$, i.e. $(x^2+y^2)^{3/2} = 4x^2$

(b) Replacing y by $-y$ leaves the equation unchanged (equivalently $r(-\theta) = r(\theta)$), so the curve is symmetric in the x-axis

(c) $r \geqslant 0$ and $r = 0$ at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$. One loop is for $-\frac{\pi}{2} \leqslant \theta \leqslant \frac{\pi}{2}$: $A = \frac{1}{2}\int_{-\pi/2}^{\pi/2} 4(1+\cos 2\theta)^2 \mathrm{d}\theta = 2\int \left(\frac{3}{2} + 2\cos 2\theta + \frac{1}{2}\cos 4\theta\right) \mathrm{d}\theta = 2\left[\frac{3}{2}\theta\right]_{-\pi/2}^{\pi/2} = 3\pi$

Question 3

(a) $4\cosh\theta = 6$ gives $\cosh\theta = 1.5$, $\theta_0 = \cosh^{-1} 1.5 = \ln\dfrac{3+\sqrt{5}}{2} = 0.9624$

(b) Area $= \frac{1}{2}\int_{-\theta_0}^{\theta_0} \left(36 - 16\cosh^2\theta\right) \mathrm{d}\theta = \int_0^{\theta_0} (36 - 8 - 8\cosh 2\theta)\, \mathrm{d}\theta = [28\theta - 4\sinh 2\theta]_0^{\theta_0} = 28\theta_0 - 8\sinh\theta_0\cosh\theta_0$.
With $\cosh\theta_0 = 1.5$, $\sinh\theta_0 = \frac{\sqrt{5}}{2}$: $= 28\theta_0 - 6\sqrt{5} = 13.53$ m^2. $13.53 \div 0.5 = 27.1$, so at least 28 tins

Question 4

(a) $A_k = \frac{1}{2}\int_0^{2\pi} \left(k + \frac{6}{k} + k\cos\theta\right)^2 \mathrm{d}\theta = \frac{1}{2}\left[\left(k+\frac{6}{k}\right)^2 2\pi + k^2\pi\right] = \pi\left(k+\frac{6}{k}\right)^2 + \frac{1}{2}\pi k^2 = \pi\left(\frac{3}{2}k^2 + 12 + \frac{36}{k^2}\right)$

(b) $\dfrac{\mathrm{d}A}{\mathrm{d}k} = \pi\left(3k - \dfrac{72}{k^3}\right) = 0$ gives $k^4 = 24$, $k = 24^{1/4}$. $\dfrac{\mathrm{d}^2A}{\mathrm{d}k^2} = \pi\left(3 + \dfrac{216}{k^4}\right) > 0$ so it is a minimum. $k^2 = 2\sqrt{6}$:
$A_{\min} = \pi\left(3\sqrt{6} + 12 + \dfrac{36}{2\sqrt{6}}\right) = \pi\left(12 + 6\sqrt{6}\right)$